

# INTRODUCTION TO DATA SCIENCE AND SYSTEMS

## CRIB SHEET

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### I. LINEAR ALGEBRA

For vectors  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n, a \in \mathbb{R}$ :

- Inner Product:  $\mathbf{x} \cdot \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$
- Orthogonality:  $\mathbf{x} \perp \mathbf{y} \iff \mathbf{x} \cdot \mathbf{y} = 0$
- Cosine Similarity:  $\cos \theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$

The general case  $p$ -**Norm** of a **vector norm** is defined as:

$$\|\mathbf{x}\|_p = \left( \sum_{i=1}^n |x_i|^p \right)^{1/p}$$

**Data Statistics** ( $\mathbf{X} \in \mathbb{R}^{m \times n}$ ,  $m$  observations,  $n$  features)

- Mean Vector:  $\boldsymbol{\mu} = \frac{1}{m} \sum_{i=1}^m \mathbf{x}_i$
- Centered Data:  $\mathbf{X}_{centered} = \mathbf{X} - \boldsymbol{\mu}$
- Covariance Matrix ( $\boldsymbol{\Sigma}$ ):

$$\boldsymbol{\Sigma} \approx \frac{1}{N-1} \mathbf{X}_{centered}^T \mathbf{X}_{centered}$$

(Symmetric)

**Singularity:** A square matrix  $\mathbf{A}$  is **singular** (non-invertible,  $\mathbf{A}^{-1}$  does not exist) if:

- $\det(\mathbf{A}) = 0$ .
- The columns (or rows) are linearly dependent.

**Eigendecomposition** (for  $\mathbf{A}$  being a **square** matrix)

- Definition:  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$  ( $\mathbf{v}$ : eigenvector,  $\lambda$ : eigenvalue)
- Decomposition (Symmetric  $\mathbf{A}$ ):  $\mathbf{A} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^T$
- $\mathbf{V}$ : Orthonormal matrix of eigenvectors.
- $\boldsymbol{\Lambda}$ : Diagonal matrix of eigenvalues.

**Singular Value Decomposition (SVD)**

$$\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$$

where:

- $\mathbf{U} \in \mathbb{R}^{m \times m}$ : Left singular vectors (orthogonal).
- $\boldsymbol{\Sigma} \in \mathbb{R}^{m \times n}$ : Diagonal matrix of singular values  $\sigma_i \geq 0$ .
- $\mathbf{V} \in \mathbb{R}^{n \times n}$ : Right singular vectors (orthogonal).
- The **pseudo-inverse** of a matrix is:  $\mathbf{A}^\dagger = \mathbf{V}\boldsymbol{\Sigma}^\dagger\mathbf{U}^T$

**Whitening a Dataset** (Zero mean, Identity Covariance  $\mathbf{I}$ )

- Centered Data SVD:  $\mathbf{X}_{centered} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$
- Whitening Transform:

$$\mathbf{X}_{whitened} = \mathbf{U}\boldsymbol{\Sigma}^{-1/2}\mathbf{U}^T\mathbf{X}_{centered}$$

- $\boldsymbol{\Sigma}^{-1/2}$  is the diagonal matrix with elements  $\sigma_i^{-1/2}$ .

### II. OPTIMISATION

If  $f(x)$  is a scalar function of a scalar  $x$

- $f'(x)$  is the first derivative of  $f$  w.r.t.  $x$ ,  $\frac{d}{dx}f(x)$ .
- The second derivative is written  $f''(x) = \frac{d^2}{dx^2}f(x)$ .

Differentiation of  $f(x)$  with **finite differences**:

$$\frac{d}{dx}f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{2h}$$

For a vector function  $\mathbf{y} = f(\mathbf{x})$  the **Jacobian matrix** is

$$f'(\mathbf{x}) = \mathbf{J} = \begin{bmatrix} \frac{\partial y_0}{\partial x_0} & \frac{\partial y_0}{\partial x_1} & \cdots & \frac{\partial y_0}{\partial x_n} \\ \frac{\partial y_1}{\partial x_0} & \frac{\partial y_1}{\partial x_1} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial x_0} & \frac{\partial y_m}{\partial x_1} & \cdots & \frac{\partial y_m}{\partial x_n} \end{bmatrix}$$

The **gradient vector** of a scalar function  $L(\theta)$  of a vector  $\theta$  is:

$$\nabla L(\theta) = \left[ \frac{\partial L(\theta)}{\partial \theta_1}, \frac{\partial L(\theta)}{\partial \theta_2}, \dots, \frac{\partial L(\theta)}{\partial \theta_n} \right]$$

The **Hessian Matrix** is:

$$H(L) = \nabla \nabla L(\theta) = \nabla^2 L(\theta)$$

$$= \begin{bmatrix} \frac{\partial^2 L(\theta)}{\partial \theta_1^2} & \frac{\partial^2 L(\theta)}{\partial \theta_1 \partial \theta_2} & \frac{\partial^2 L(\theta)}{\partial \theta_1 \partial \theta_3} & \cdots & \frac{\partial^2 L(\theta)}{\partial \theta_1 \partial \theta_n} \\ \frac{\partial^2 L(\theta)}{\partial \theta_2 \partial \theta_1} & \frac{\partial^2 L(\theta)}{\partial \theta_2^2} & \frac{\partial^2 L(\theta)}{\partial \theta_2 \partial \theta_3} & \cdots & \frac{\partial^2 L(\theta)}{\partial \theta_2 \partial \theta_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 L(\theta)}{\partial \theta_n \partial \theta_1} & \frac{\partial^2 L(\theta)}{\partial \theta_n \partial \theta_2} & \frac{\partial^2 L(\theta)}{\partial \theta_n \partial \theta_3} & \cdots & \frac{\partial^2 L(\theta)}{\partial \theta_n^2} \end{bmatrix}$$

The Lipschitz constant  $K$  of a function  $f(x)$  is:

$$K = \sup \left[ \frac{|f(x) - f(y)|}{|x - y|} \right]$$

The **Least Squares** solution  $\mathbf{x}^*$  for the problem:

$$\arg \min_x L(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2,$$

is given by

$$\mathbf{x}^* = \mathbf{A}^+ \mathbf{y}$$

A **gradient descent** step optimising a loss function  $L(\theta)$  with parameters  $\theta$ :

$$\theta^{(i+1)} = \theta^{(i)} - \delta \nabla L(\theta^{(i)})$$

**Gradient descent with momentum:**

$$v = \alpha v + \delta \nabla L(\theta) \theta^{(i+1)} = \theta^{(i)} - v$$

### III. PROBABILITIES

#### Axioms of probability:

- Boundedness:  $0 \leq P(A) \leq 1$
- Unitarity:  $\sum_x P(x) = 1$
- Sum rule:  $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$
- Conditional Probability:  $P(A|B) = \frac{P(A \wedge B)}{P(B)}$

#### Integration to Unity

- continuous:  $\sum_i f_X(x_i) = 1$
- discrete:  $\int_x f_X(x) dx = 1$

#### Expectation:

- continuous  $\mathbb{E}[X] = \int_x x f_X(x) dx$
- discrete  $\mathbb{E}[X] = \sum_x x f_X(x)$

The expectation of any function  $g(X)$  of a continuous random variable  $X$  is defined as:

- continuous:  $\mathbb{E}[g(X)] = \int_x f_X(x) g(x) dx$
- discrete:  $\mathbb{E}[g(X)] = \sum_x f_X(x) g(x)$

#### Odds and logit

$$\text{odds} = \frac{p}{1-p}$$

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

#### Conditional Probability:

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}$$

#### Marginal probability:

- continuous:  $P(X) = \int_y P(X, Y) dy$
- discrete:  $P(X) = \sum_y P(X, Y)$

#### Bayes Rule:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

#### Cumulative Distribution:

$$F_X(x) = \int_{-\infty}^x f_X(x) = P(X \leq x)$$

#### Normal Distribution:

$$f_X(x) = \frac{1}{Z} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

#### Mean and Variance

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2.$$

### IV. DATABASES

#### Data Storage & I/O Cost

- **I/O Cost:** Measured by the number of block accesses (reads/writes) to disk.
- **Blocking Factor (bfr):** Number of records per disk block.
  - $bfr = \left\lfloor \frac{\text{Block Size}}{\text{Record Size}} \right\rfloor$
- **Total Blocks (b):** Total number of blocks for a relation.
  - $b = \left\lceil \frac{\text{Number of Records } (r)}{\text{Blocking Factor } (bfr)} \right\rceil$
- **Search Costs (Block Accesses / Complexity):**
  - **Linear Search (Heap/Unsorted):**  $O(b)$
  - **Binary Search (Sequential/Sorted):**  $O(\log_2(b))$
  - **Index-based Search (B+ Tree):**  $O(\log_{bfr_1}(b))$  or  $O(t)$ , where  $t$  is the number of levels.

#### Indexing Concepts

- **Primary Index:** Index on the **ordering key** field. Always **Sparse** (one entry per data block).
- **Clustering Index:** Index on an **ordering non-key** field. Always **Sparse** (one entry per distinct value/cluster).
- **Secondary Index:** Index on a **non-ordering** field. Always **Dense** (one entry per data record).
- **Multi-Level Index (Sparse):** Used when the first-level index is too large to fit in one block.
  - **Number of Entries at L1:** Equals the number of data blocks,  $b$ .
  - **Index Block Factor (bfr<sub>i</sub>):**  $\left\lfloor \frac{\text{Block Size}}{\text{Key Size} + \text{Pointer Size}} \right\rfloor$
  - **Search Cost:**  $t + 1$ , where  $t$  is the number of index levels.

#### Query Processing & Optimisation

- **Heuristic Optimization:** Applies relational algebra rules. Key rule: **Push selections ( $\sigma$ ) down** the plan tree to reduce intermediate result sizes early.
- **Cost-based Optimisation:** Chooses the plan with the **minimum cost** based on statistical estimates.
- **Selectivity (sl):** Fraction of tuples satisfying a condition ( $\in [0, 1]$ ).
- **Selection Cardinality (s):** Expected number of result tuples.  $s = r \cdot sl$ .

#### Join Algorithms (BNLJ)

- **Block Nested-Loop Join (BNLJ):** Efficient disk-based join using  $M$  buffer blocks.
- **Cost (General):**  $\text{Cost} = b_{\text{Outer}} + \left( \left\lceil \frac{b_{\text{Outer}}}{M-2} \right\rceil \times b_{\text{Inner}} \right)$ .

#### Specialized Data Systems

- **Key-Value, Document, and Columnar Databases:** Used for high-throughput, flexible schema, or analytical workloads, offering greater scalability and availability than RDBMS for non-relational data patterns.
- **Graph and Vector Databases:** Specialized for handling complex relationships or high-dimensional data. They support unique query types like pathfinding, approximate nearest neighbor (ANN) search, etc.